

Digital Pendulum Installation & Commissioning

33-936IC

Feedback Instruments Ltd., Park Road, Crowborough, East Sussex, TN6 2QX, UK

Telephone: +44 (0) 1892 653322, Fax: +44 (0) 1892 663719

E-mail: feedback@feedback-instruments.com

website: <http://www.feedback-instruments.com>

Manual: 33-936IC Ed03 082013

Feedback Part No. 1160-33936IC

THE HEALTH AND SAFETY AT WORK ACT 1974

We are required under the Health and Safety at Work Act 1974, to make available to users of this equipment certain information regarding its safe use.

The equipment, when used in normal or prescribed applications within the parameters set for its mechanical and electrical performance, should not cause any danger or hazard to health or safety if normal engineering practices are observed and they are used in accordance with the instructions supplied.

If, in specific cases, circumstances exist in which a potential hazard may be brought about by careless or improper use, these will be pointed out and the necessary precautions emphasised.

While we provide the fullest possible user information relating to the proper use of this equipment, if there is any doubt whatsoever about any aspect, the user should contact the Product Safety Officer at Feedback Instruments Limited, Crowborough.

This equipment should not be used by inexperienced users unless they are under supervision.

We are required by European Directives to indicate on our equipment panels certain areas and warnings that require attention by the user. These have been indicated in the specified way by yellow labels with black printing, the meaning of any labels that may be fixed to the instrument are shown below:



CAUTION -
RISK OF
DANGER



CAUTION -
RISK OF
ELECTRIC SHOCK



CAUTION -
ELECTROSTATIC
SENSITIVE DEVICE

Refer to accompanying documents

PRODUCT IMPROVEMENTS

We maintain a policy of continuous product improvement by incorporating the latest developments and components into our equipment, even up to the time of dispatch.

All major changes are incorporated into up-dated editions of our manuals and this manual was believed to be correct at the time of printing. However, some product changes which do not affect the instructional capability of the equipment, may not be included until it is necessary to incorporate other significant changes.

COMPONENT REPLACEMENT

Where components are of a 'Safety Critical' nature, i.e. all components involved with the supply or carrying of voltages at supply potential or higher, these must be replaced with components of equal international safety approval in order to maintain full equipment safety.

In order to maintain compliance with international directives, all replacement components should be identical to those originally supplied.

Any component may be ordered direct from Feedback or its agents by quoting the following information:

- | | |
|------------------------|----------------------------|
| 1. Equipment type | 3. Component value |
| 2. Component reference | 4. Equipment serial number |

Components can often be replaced by alternatives available locally, however we cannot therefore guarantee continued performance either to published specification or compliance with international standards.

CE DECLARATION CONCERNING ELECTROMAGNETIC COMPATIBILITY

Should this equipment be used outside the classroom, laboratory study area or similar such place for which it is designed and sold then Feedback Instruments Ltd hereby states that conformity with the protection requirements of the European Community Electromagnetic Compatibility Directive (89/336/EEC) may be invalidated and could lead to prosecution.

This equipment, when operated in accordance with the supplied documentation, does not cause electromagnetic disturbance outside its immediate electromagnetic environment.

COPYRIGHT NOTICE

© Feedback Instruments Limited

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior permission of Feedback Instruments Limited.

ACKNOWLEDGEMENTS

Feedback Instruments Ltd acknowledge all trademarks.

MATLAB is a registered trademark of Mathworks Inc.

Table of Contents

Manual overview.....	2
Required manuals and equipment	3
Mechanical assembly	5
Advantech PCI1711U card installation guide.....	8
Installing from a CD – new installation	8
Upgrading and installing downloaded files	13
Pendulum software installation	14
Location of Simulink model files	19
Pendulum Mechanical Unit description	21
Control system.....	22
Computer control	23
Running a Simulink Pendulum controller.....	25
Testing the real-time control system.....	25
<i>Adding an I/O card to the Matlab installed boards list.....</i>	<i>27</i>
<i>Running a real-time model</i>	<i>28</i>
Designing controllers	28
Reading the encoders and setting the control signal	29
<i>PCI-1711U I/O Block – Feedback Encoder.....</i>	<i>29</i>
<i>PCI 1711U I/O Block – Feedback DAC.....</i>	<i>32</i>
<i>Pendulum signals</i>	<i>34</i>
Troubleshooting.....	35
Controller runs weakly, cart motor seems to have little power	35
Masks have no background and show question marks.....	37
Project files corrupted after crash during simulation or build stage	37
Incorrect kernel version installed, when running a real-time simulation.....	38
Unresponsive Windows, freeze or crash on simulation.....	38

Manual overview

This manual explains how to install and test the pendulum equipment. The basic steps described are: assembly and adjustment of the mechanical unit, installation of the PC interface card and its software drivers, installation of the Pendulum control software, interconnection and basic system testing. An overview of how the pendulum is interfaced and controlled with the PC and Matlab environment is given to help the user produce his own controller design.

Required manuals and equipment

The following equipment and manuals are required for the use of the Digital Pendulum Control system:

- 1 PC with Windows 7 SP1 or Windows 10, all 64 bit and a free PCI slot on the motherboard.
- 2 Matlab R2015b (V8.6) or later with Simulink on Windows 7 Sp1, or Matlab Version R2017a (V9.2) or later with Simulink on Windows 10.
- 3 Matlab Toolboxes:
 - Real Time Workshop with Simulink Desktop Real-Time Windows Target
 - System Identification Toolbox (necessary to perform identification exercises)
 - Control System Toolbox (necessary to perform controller synthesis)

The Simulink Desktop Real-Time Target must be activated by opening the Matlab Command Window and typing:

`>> sldrtkernel -setup` (note the "space" before the "-")

Press the "enter" key to execute the above command.

- 4 Advantech PCI-1711U card with drivers
- 5 33-005 Digital Pendulum Control System – Mechanical Unit and Digital Controller
- 6 Feedback 33-936 software installation CDs
- 7 'Installation & Commissioning' manual
- 8 'Control Experiments' manual
- 9 'Matlab Guide' manual

The following equipment is supplied by Feedback Instruments:

1 33-200 - Digital Pendulum Mechanical Unit

2 33-201 - Digital Pendulum Controller

and either:

3 33-936 - Digital Pendulum Control System Software including Advantech PCI-1711U card

or

4 33-936I - Digital Pendulum Control System Software excluding Advantech PCI-1711U card

Composite products:

1 33-005 – Includes: 33-200, 33-201 and 33-936

2 33-005I – Includes: 33-200, 33-201 and 33-936I

Mechanical assembly

Unpack the items and remove the securing material. You should have the following items:

- 1 Track for the cart to move on, with motor and toothed belt
- 2 Two legs with adjustable feet to support the track
- 3 Pendulum cart with mounting points for the pendulum arms and an attached ribbon cable
- 4 Two pendulum arms with weights attached
- 5 1 x 2mm and 1 x 6mm Allen keys

Figure 1 shows the assembled Digital Pendulum Mechanical Unit.

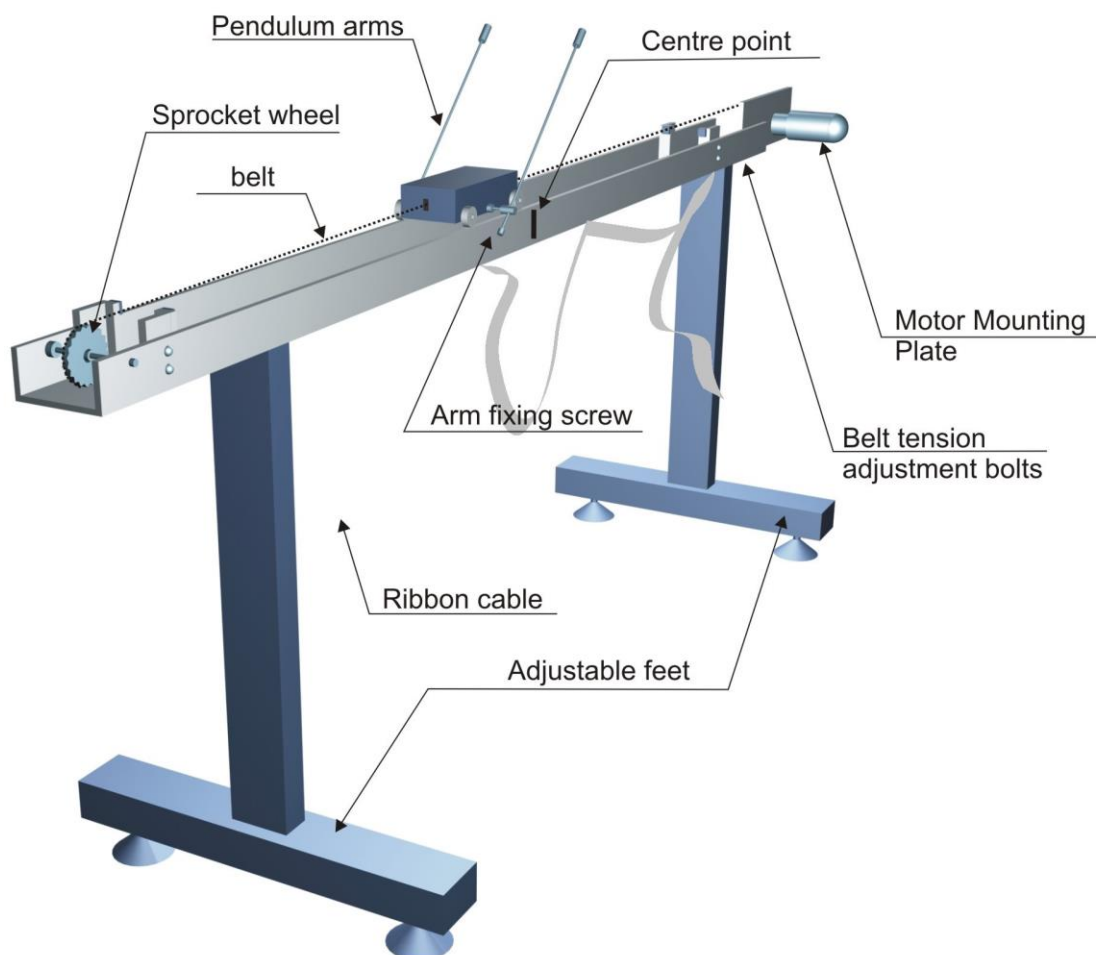


Figure 1 Digital Pendulum Mechanical Unit

Place the Digital Pendulum Mechanical Unit on a level surface with the motor on the left when viewed from the front and allow sufficient room for the controller box and the PC and mouse. Ensure that the pendulum arms are clear of any obstructions when executing a complete revolution, and are not able to make contact with any observers.

Using the 6 mm Allen key provided, remove the 8 bolts attached to the track at the leg mounting positions and fix the legs to the track with these bolts. Tighten the bolts securely.

Use the adjustable feet (it is only necessary to adjust one of the four) to ensure that the pendulum system is stable on the surface. Slacken off the adjustable foot locking nut and rotate the foot until the system is perfectly stable. Tighten the locking nut.

Insert one of the pendulum arms into one of the two mounting positions on the cart from the bottom and loosely tighten the securing screw on to the flat part of the pendulum arm with the 2mm Allen key provided. Push the arm upwards until the end of the flat is reached and tighten the securing screw. Repeat for the other pendulum arm on the other side of the cart.

Ensure that the pendulum cart is free to move on the track and that the drive belt is attached to the cart at both ends and goes over the sprocket wheels at each end of the track. If the belt is slack, use the 6mm Allen key to slacken off the belt tension adjustment bolts and move the motor mounting plate until the belt is taut. Tighten the adjustment bolts.

Finally, clip the ribbon cable to the pendulum left hand leg using the attached clips.

Advantech PCI1711U card installation guide

It is assumed that a PC is available with Matlab, Simulink, Real-Time Workshop and Simulink Desktop Real-Time Kernel correctly installed and functioning.

Make sure you follow the instructions given in this section during the installation of the Advantech PCI-1711U card. This will ensure that you avoid common installation problems.

You must have administrator privileges on the PC before you can carry out this software installation.

Install the software supplied with the PCI-1711U card first. The PCI-1711U card must not be built in until the software has been installed correctly. A PCI-1711U device driver of version 3.1.12 or higher is recommended.

Installing from a CD – new installation

Ensure that you have the latest driver (DAQNavi Driver for PCI-1711U). If not, download the files from the Advantech website and refer to the procedure above for installation.

1. Insert the CD supplied with the PCI-1711U card into the PC. The installation program will automatically start. The following window will open (Figure 2).

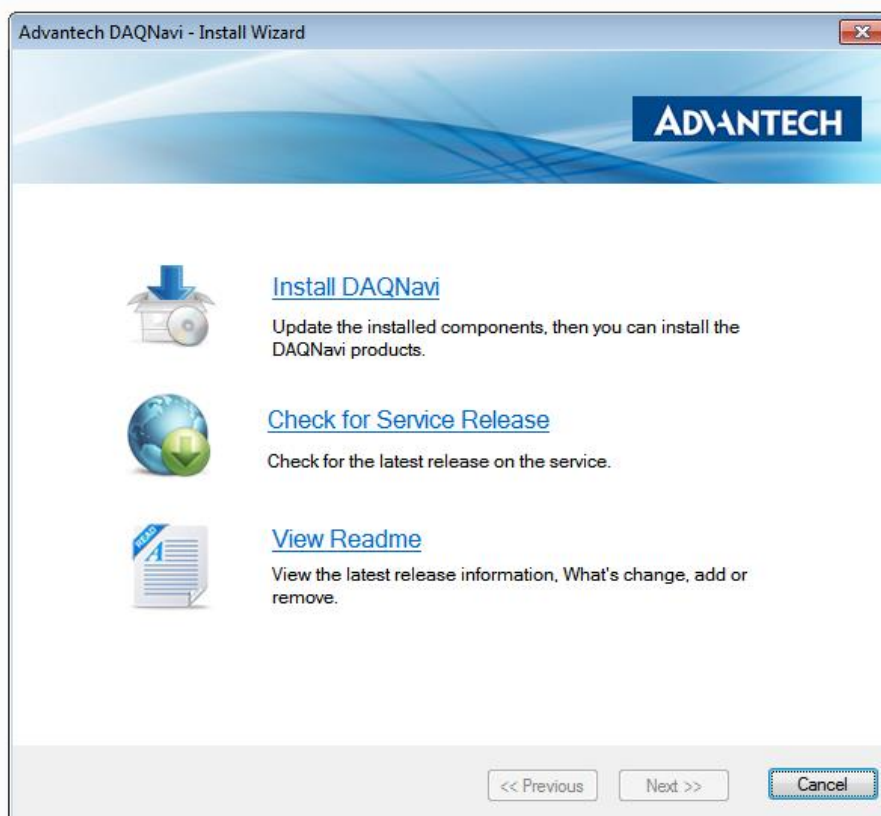


Figure 2 Initial Installation Window

2. To install the software select 'Install DAQNav'. The following window will appear (Figure 3)

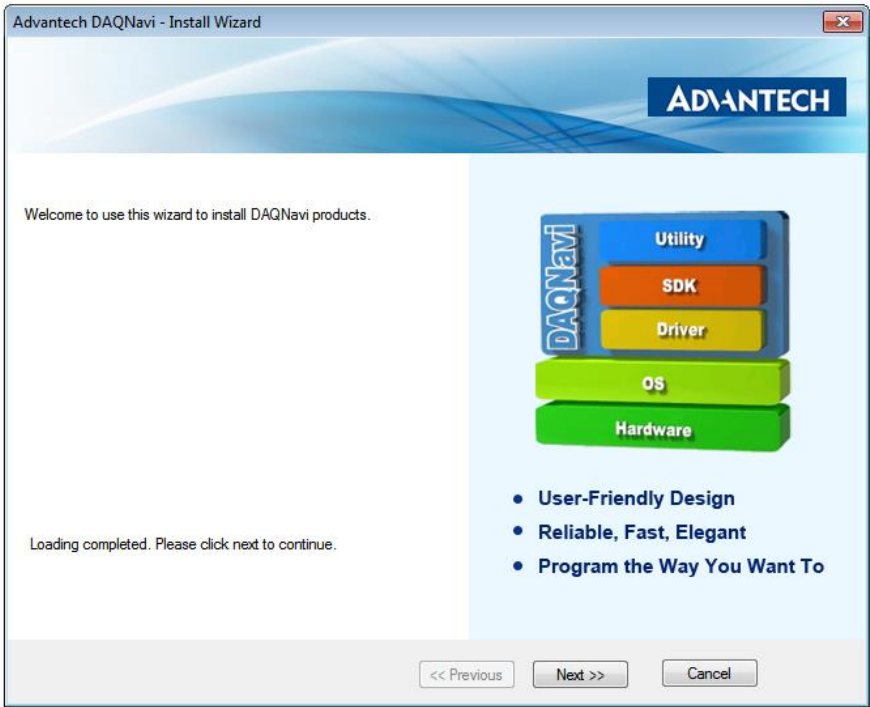


Figure 3 Installation Element Selection Window

3. Press 'Next' to continue to the User information window (Figure 4).

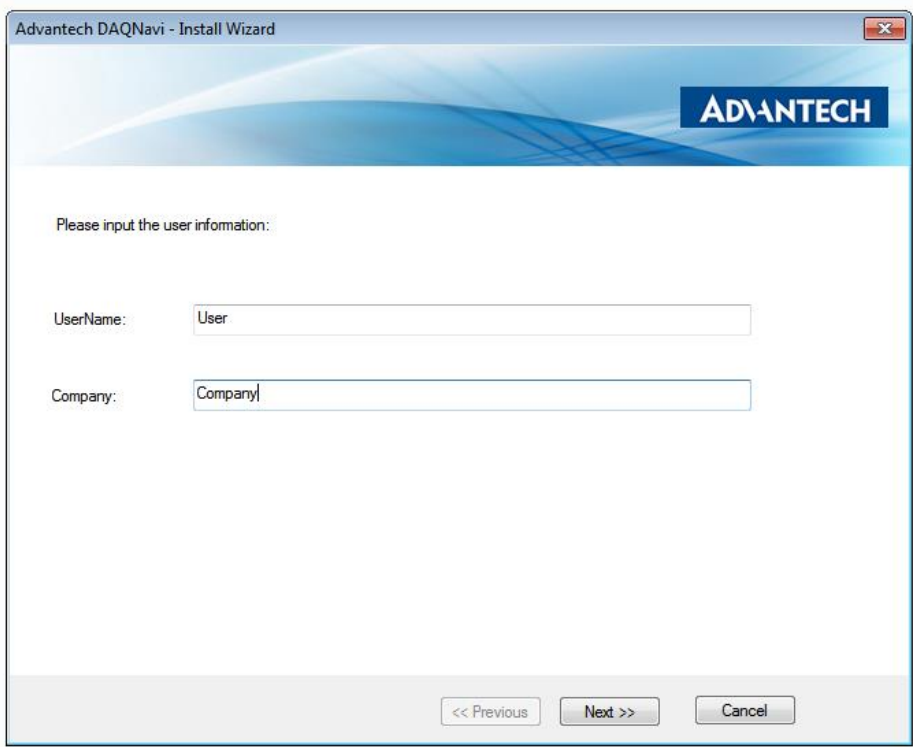


Figure 4 User Information Window

4. Fill the fields with proper information and press 'Next'. The installation options window will appear (Figure 5).

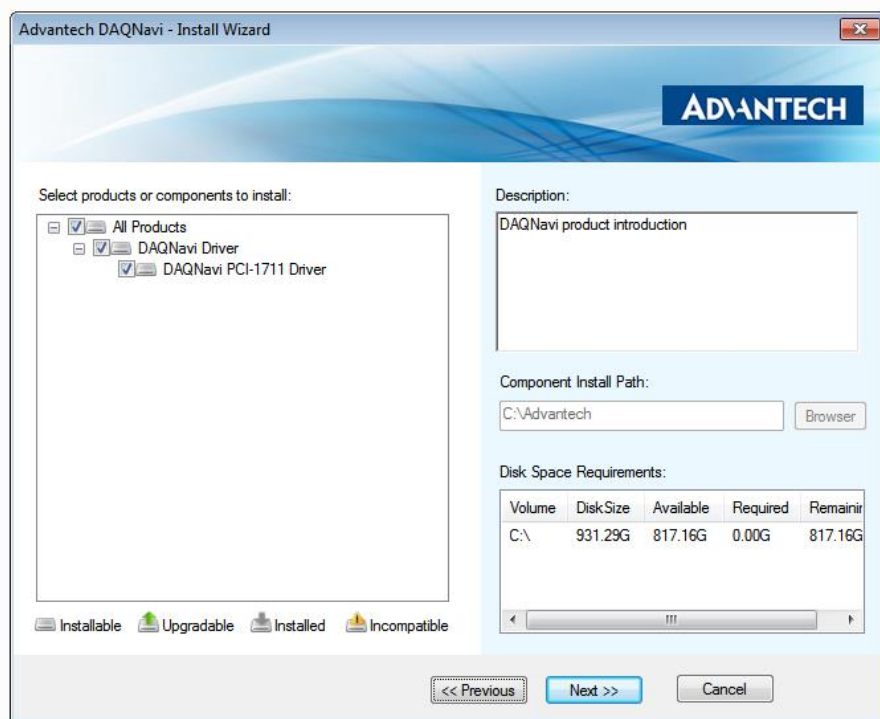


Figure 5 Destination Folder Window

5. Choose the install path, make the driver component is checked and press 'Next'. A prompt from Windows will appear (Figure 6).

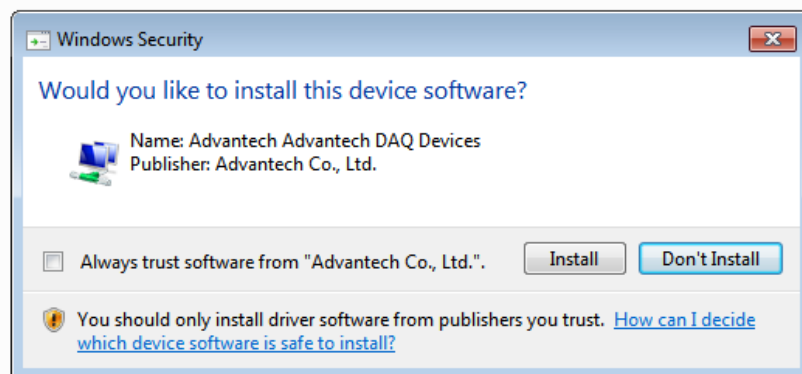


Figure 6 Setup Type Window

6. Press 'Install' and the installation will continue (Figure 7).

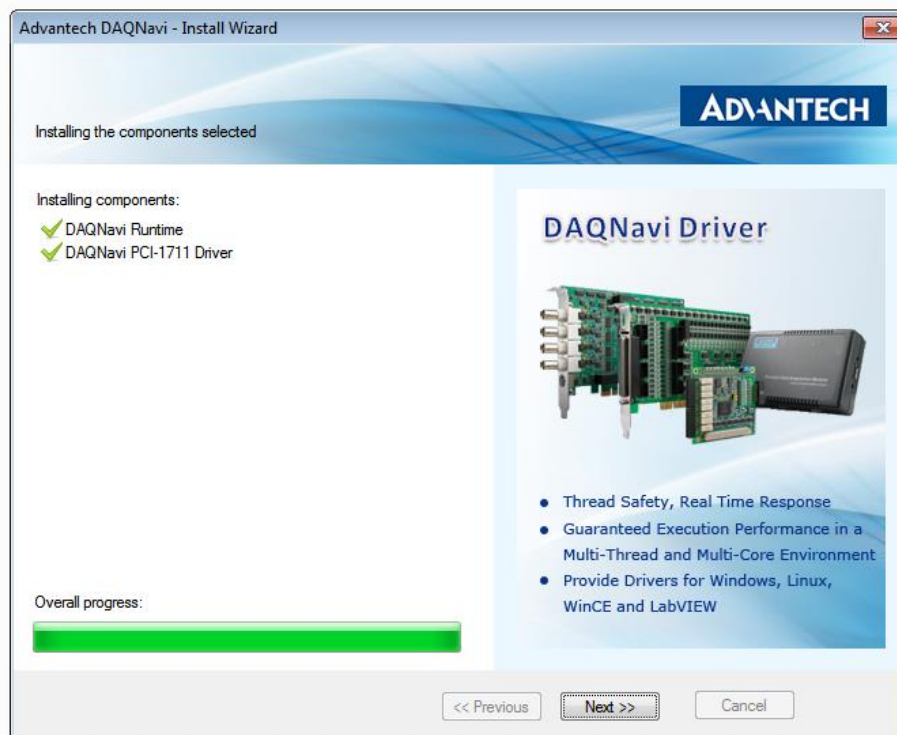


Figure 7 Program Folder Selection Window

7. Once the progress bar is full, press next. Press finish on the next window and the installation for the driver is complete.
8. Now shut down the computer and insert the PCI Card into the computer. Ask the responsible IT technician for help, if you are unsure how to do that. There is a possibility to damage the computer and/or the card, if you handle them inappropriately.

Installation is now complete. To check that the card has installed correctly, open the Windows Device Manager by holding the windows button and pressing R, entering devmgmt.msc and pressing ok. There should be an entry called Advantech DAQ Devices with the PCI 1711U Device in it.

Upgrading and installing downloaded files

The latest version of software can be downloaded from the Advantech website if you have an older version. You will need to download and install the DAQNav Driver for PCI-1711U

First uninstall any previous version of the drivers:

Uninstall Advantech DAQNav Products either using uninstall from the Advantech Program Menu entry or from Control Panel - Add/Remove programs.

Navigate to C:\Program Files and delete the Advantech subdirectory and all its contents.

Do NOT reboot the PC.

Now run the driver installation as described in the previous section.

Reboot the PC.

Installation is now complete. To check that the card has installed correctly, open the Windows Device Manager by holding the windows button and pressing R, entering "devmgmt.msc" and pressing ok. There should be an entry called Advantech DAQ Devices with the PCI 1711U Device in it.

Pendulum software installation

After installing the Advantech software and PCI-1711U card you are ready to install the Pendulum workshop software.

1. Close Matlab. Run the fbsetup program from the CD. The following window will open (Figure 8).



Figure 8 Installation window

2. If you accept the agreement, select the appropriate option and click Next.
3. After launching the setup program, the program installation manager dialog will appear on the screen (Figure 9).

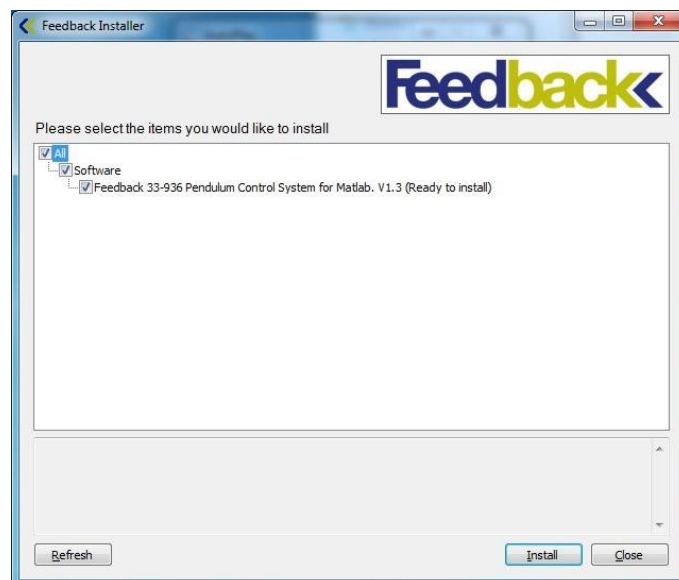


Figure 9 Program Installation Manager window

Tick the software to be installed. Click the 'Install' button.

4. The Matlab version choice window will appear (Figure 10).

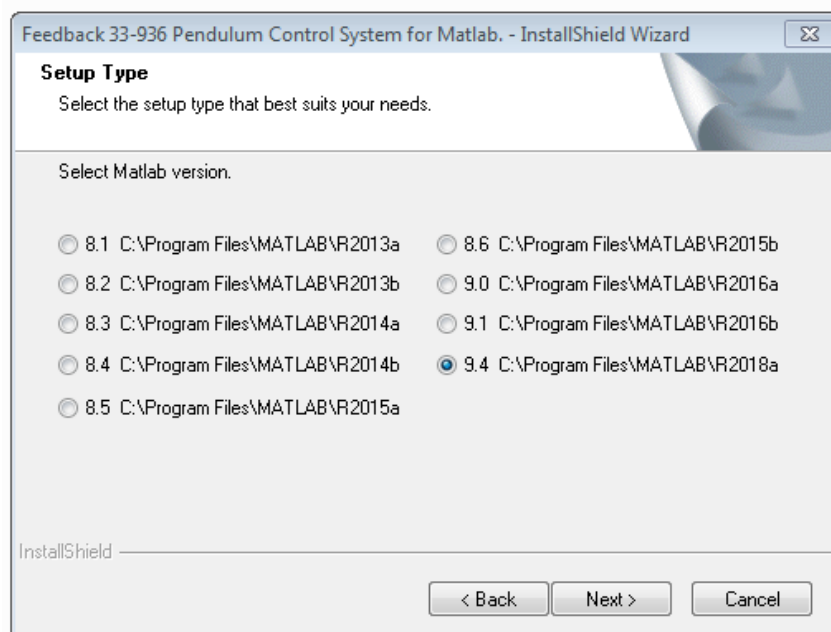


Figure 10 Matlab version choice window

5. Select the Matlab version and click 'Next' to proceed.

6. The software will begin installing. When install an 'Installation Complete' window will appear (Figure 11).

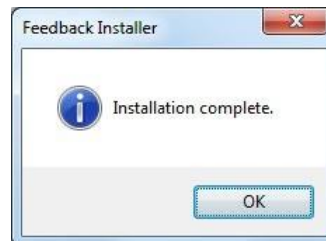


Figure 11 Installation Complete window

7. When software is installed the 'Program Installation Manager' will show the software is installed (Figure 12). Click 'Close' the installation manager.

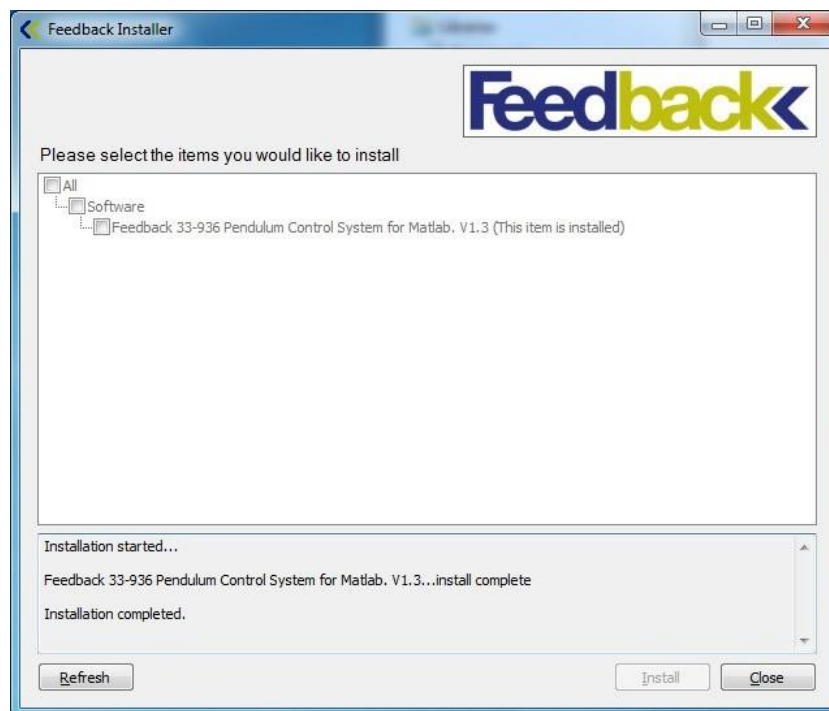


Figure 12 Program Installation Manager window

8. Two folders will be available after the installation. 'Real-time models' folder in which the Pendulum Real Time applications are loaded and 'Simulation models' folder in which all of the model simulations files are placed.

PC and pendulum connection

With all of the software installed you are ready to finish the connections of the hardware. Four elements have to be connected to complete the setup:

1. PC with the PCI-1711U card
2. Feedback SCSI Cable Adaptor
3. Digital Pendulum Controller
4. Digital Pendulum Mechanical Unit

Refer to the connection schematic presented in Figure 13 and follow the steps below.

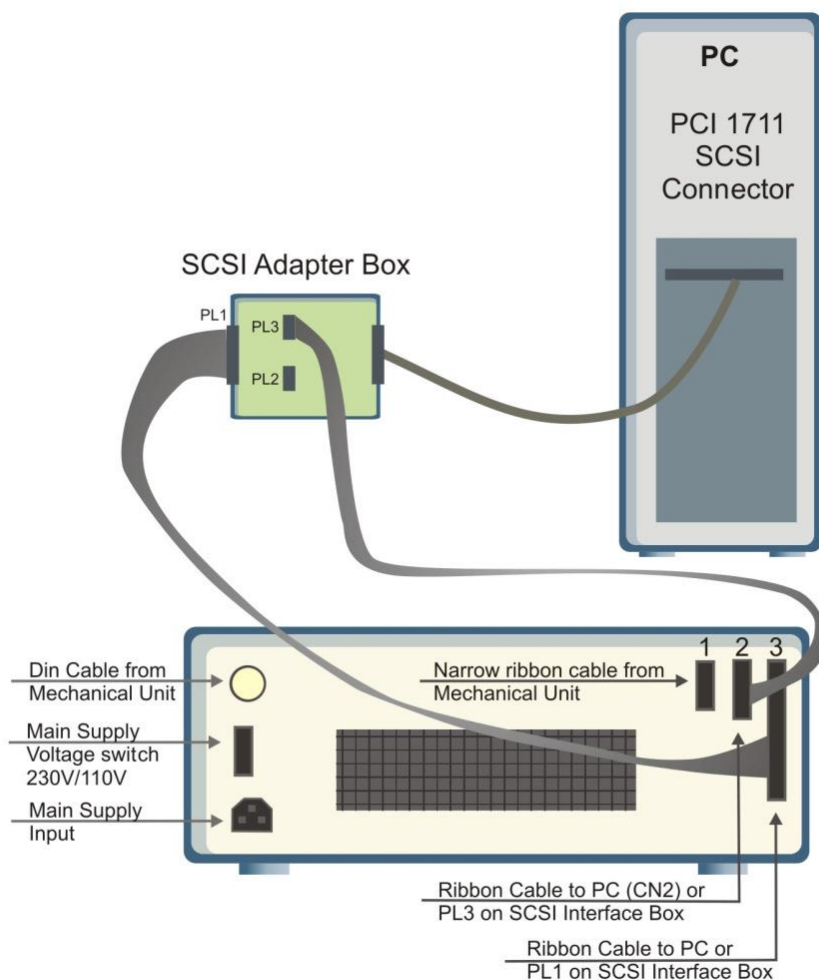


Figure 13 Connection diagram

1. Connect one end of the SCSI cable into the PCI-1711U SCSI connector at the rear of the PC. Connect the other end of the cable into the Feedback SCSI Cable Adaptor box.
2. Connect one end of the wide ribbon cable to the Cable Adaptor box PL1 port and the other end to the Digital Pendulum Controller Unit port 3.
3. Connect one end of the medium sized ribbon cable to the Cable Adaptor box PL3 port and the other end to the Digital Pendulum Controller Unit port 2.
4. The Pendulum Mechanical Unit is equipped with a narrow ribbon cable. Insert the free end of the cable into port 1 of the Digital Pendulum Controller Unit port 1.
5. The Pendulum Mechanical Unit is equipped with a circular 3-pin DIN connector. Insert the DIN connector into the Digital Controller DIN port. Connect the green earth cable to the tab provided below the DIN port.
6. Make sure the Digital Pendulum Controller **power switch is turned off**. Check that the voltage selector is set correctly for your area. Connect the unit to the mains supply using the IEC power cable.

WARNINGS

Do not switch on or press the green start button of the digital controller unit yet. Before you turn the controller on you must be sure that the PCI-1711U card is initialized correctly.

It is safe to press the green start controller button only when a Simulink control application was connected to the external interface and running at least once since the computer has been turned on.

Make sure that all of the pendulum mechanical unit bolts are tightened securely.

Always start the simulation with the pendulum placed in the middle of the rail.

Keep clear of the working pendulum and ensure that it will not accidentally come into contact with other objects

Location of Simulink model files

The example Simulink models supplied are stored in two subdirectories. Their location can be selected during installation. The default locations are:

C:\ProgramData\Feedback Instruments\33-936 Pendulum\Examples\Real-time models
C:\ProgramData\Feedback Instruments\33-936 Pendulum\Examples\Simulation models

These models can be edited, saved, built etc. as required. A protected copy of the example models is saved so that these copies can be used to reinstate the original models when required. The default locations of the protected examples are:

C:\ProgramData\Feedback Instruments\33-936 Pendulum\Example backup\Real-time models
C:\ProgramData\Feedback Instruments\33-936 Pendulum\Example backup\Simulation models

To reinstate the original models, delete all files within the *Examples\Real-time models* and *Examples\Simulation models* directories and copy in the contents of the *Example backup\Real-time models* and *Example backup\Simulation models* directories respectively.

Load Matlab and the menu system using the Windows Start menu or by double clicking on the desktop icon (Figure 14).

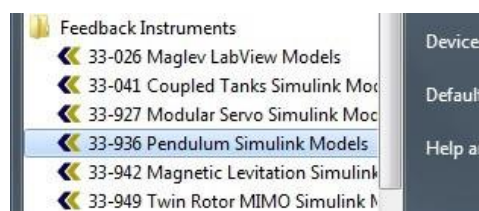


Figure 14 Start menu

Matlab will run and the Simulink model menu will open (Figure 15).

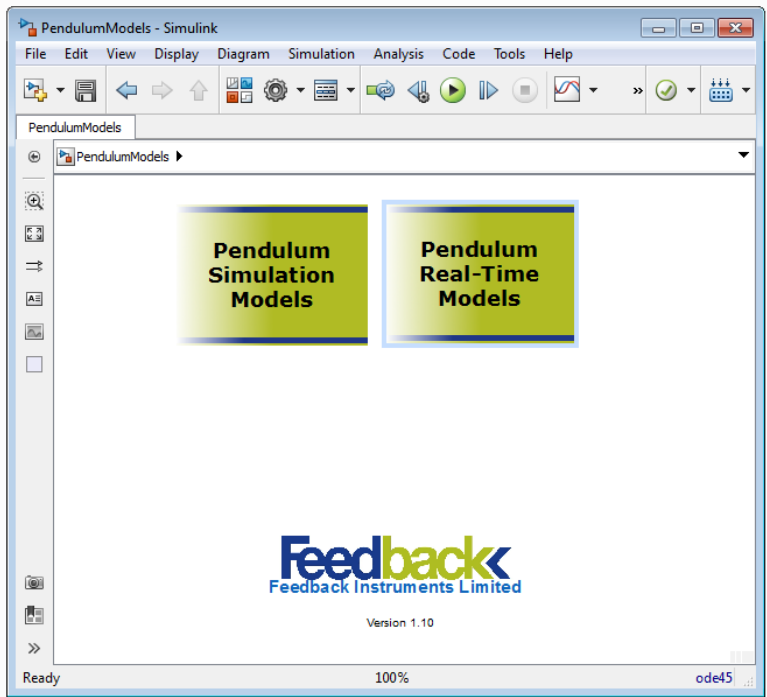


Figure 15 Simulink model menu

To run a real-time model for example, double click on the 'Pendulum Real-Time Models' block in the main menu. A sub-menu containing all of the real-time models will open. Double click the required model, for example Cart Control. The Simulink model window will open.

Refer to '*Running a Simulink Pendulum controller*' section for more information if necessary.

Pendulum Mechanical Unit description

The main element of the Pendulum Mechanical Unit is a cart with two pendulums travelling on a rail. The cart position is controlled by the toothed belt which is driven by a DC motor with a velocity proportional to the control voltage applied to the motor (Figure 16).

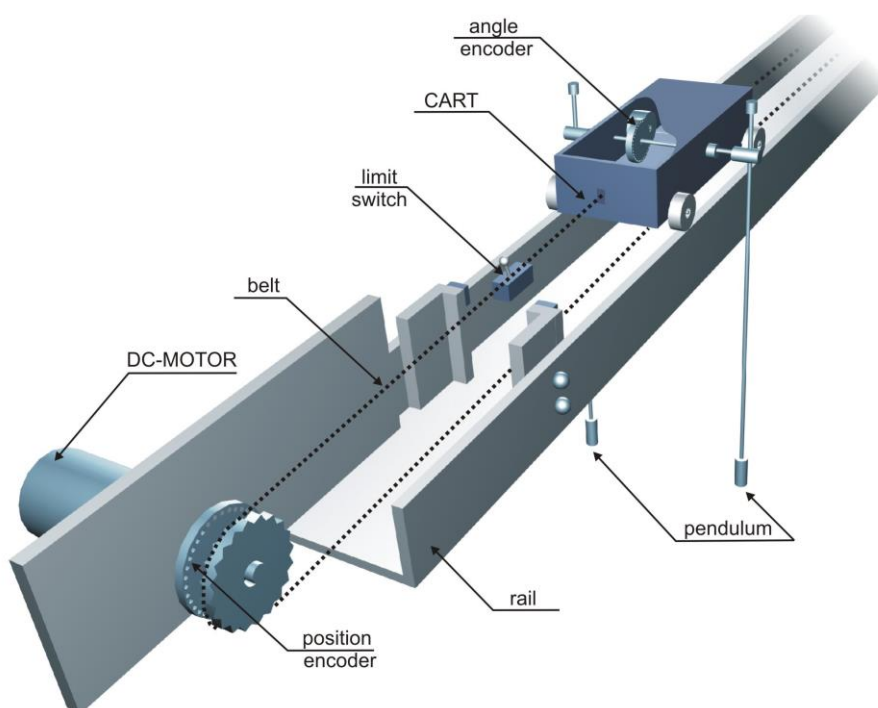


Figure 16 Pendulum Mechanical Unit

The movement of the cart on the rail is bounded mechanically. Additionally, safety limit switches are mounted at either end of the track which cut power to the motor when the cart crosses them.

The cart relative position and relative pendulum angle can be measured by the optical encoders (Figure 16). An optical encoder consists of a light source, light detector and a slit disk placed between them. This way the relative position with respect to the initial point can be measured by counting the pulses on the light detector. Refer to further sections of this manual for explanation on how the encoder pulses are counted.

Because the relative position and angle are measured every time the simulation is started, the cart has to be placed in the zero position with the pendulum not moving before the simulation is started.

Control system

The Pendulum Control System is organized as shown in the schematic presented in Figure 17.

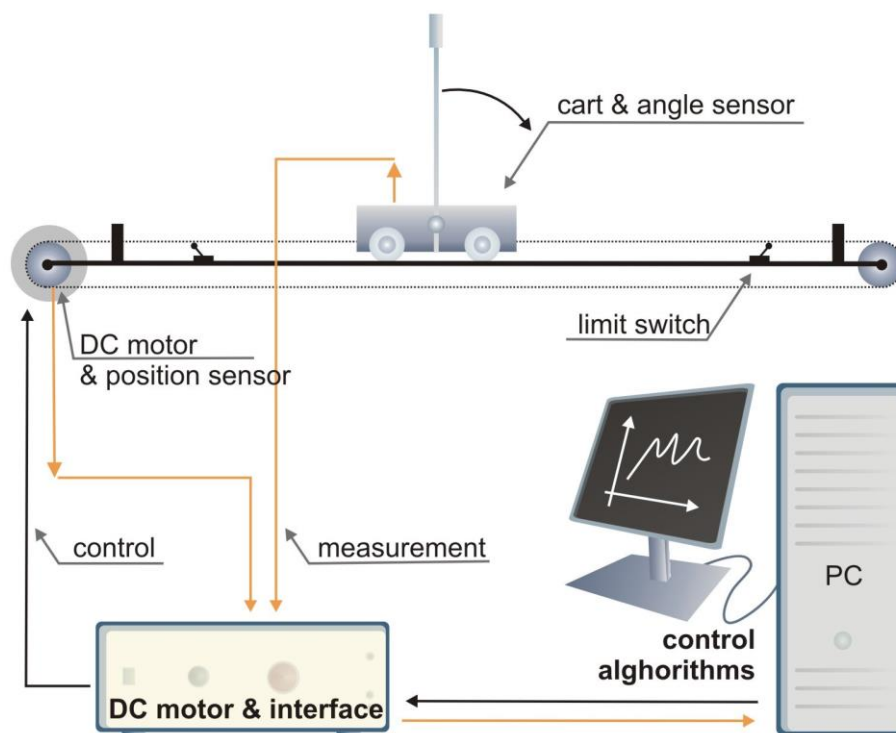


Figure 17 Control System Schematic

The PC with Advantech card and Matlab and Simulink environment serve as the main control unit. The control signal, which is a voltage between -2.5 V and 2.5 V , is transferred to the Digital Pendulum Controller, which drives the DC motor.

The cart position and the pendulum angle encoder signals are transferred to the Digital Pendulum Controller and then to the PC, where all of the control algorithms are placed in the Simulink environment. From the Simulink point of view the pendulum is seen as the following model presented in Figure 18:

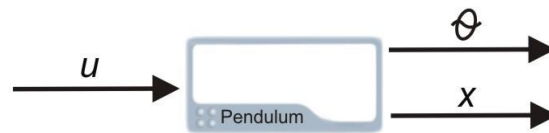


Figure 18 Pendulum model

Computer control

Advantech and Mathworks cooperation lead to significant simplification of control of the systems such as the Digital Pendulum Unit. To design a controller you do not have to build a Real Time application from the very beginning. The framework is already provided for you. You can design arbitrary controllers in the Simulink environment, execute them in real-time using the Real Time workshop and control an external process using PC I/O cards. For those not familiar with computer control systems some basic information is given here.

The digital control diagram presented in Figure 19 is used for that purpose.

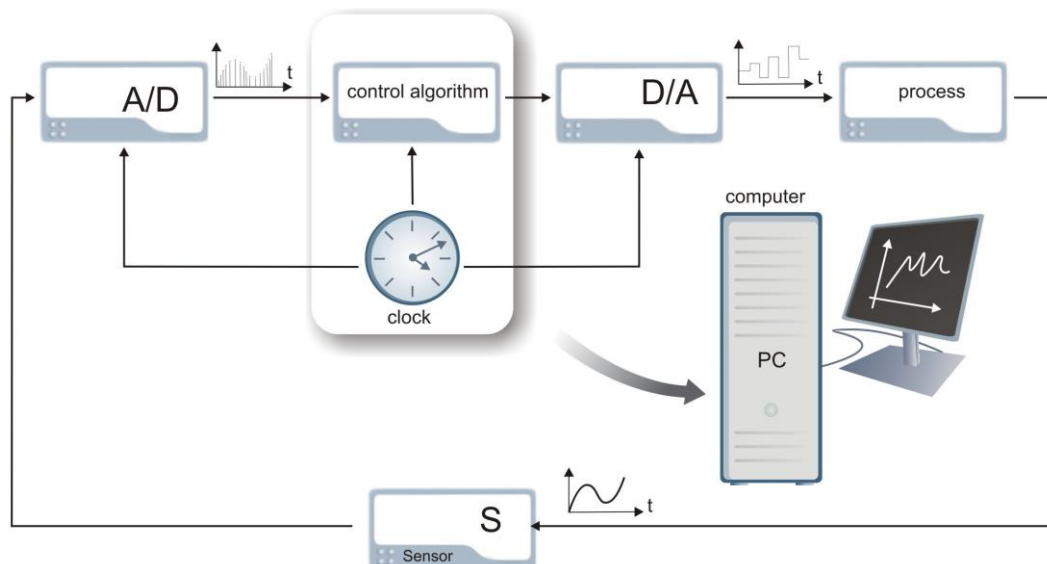


Figure 19 Computer Control System Diagram

The system consists of four main elements:

1. PC with a clocked control algorithm
2. A/D and D/A converters – serving as an interface between the PC and external environment
3. The controlled process
4. Sensors

The control algorithm and the converters operate according to the time pulses distributed by the clock. The distance between consecutive time pulses is called the sampling time. Usually the sampling time is constant however some special applications use intermittent sampling. The clock delivers an interrupt and the interrupt service routine - ISR is called during which the A/D delivers the discrete representation of the measurement returned by the sensor and based on that the algorithm calculates the value of the control signal. At the end of the ISR that value is updated and set by the D/A converter to be held for the next sampling interval.

The same control scheme is used for the Digital Pendulum System.

Running a Simulink Pendulum controller

After software installation two folders should be available, in which Simulink files are placed. One of them includes files for simulation only and the other for external Pendulum real-time control applications.

The files prepared for model simulations can be used in the same way as any normal simulation experiment conducted in Simulink.

Testing the real-time control system

The easiest way to test the newly installed system is to run one of the model files supplied. *CartControl.mdl* is suitable for this purpose (Figure 20).

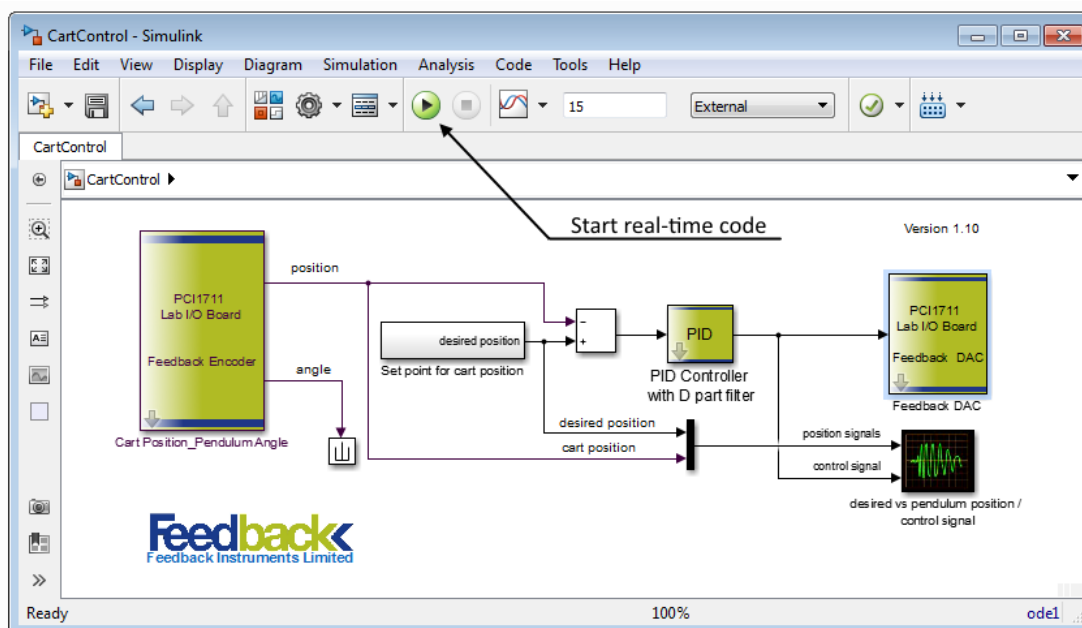


Figure 20 PID Cart Control

The models can be run by double clicking on the desktop icon or navigating using either the Windows *start* menu:

Start menu → *All Programs* → *Feedback Instruments* → *Feedback 33-936 Pendulum Simulink Models*

The top level menu will open (Figure 21).

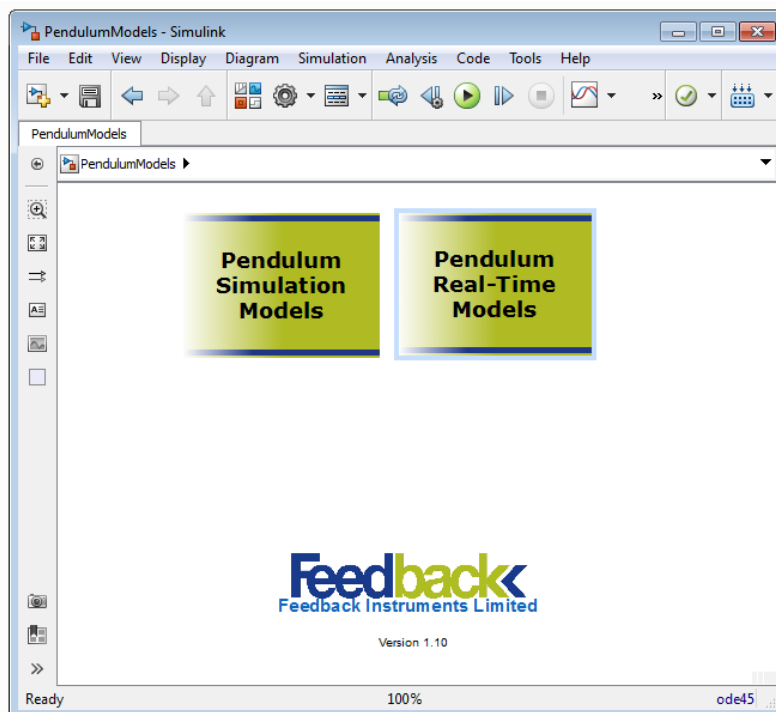


Figure 21 Model category window

To run a real-time model for example, double click on the 'Pendulum Real-Time Models' block in the main menu. A sub-menu containing all of the real-time models will open. Double click the required model, for example Cart Control. The Simulink model window will open (Figure 20).

The simulation parameters are already set. If you want to make your own applications you have to use the same simulation parameters like solver options for example. Thus it is easiest to use one of the existing applications and modify it as necessary without changing the simulation parameters. Make sure to make a copy instead of overwriting the original files. If that happens you can always find the original models in the examples backup folder.

Refer to the following sections to see how the model can be compiled and run.

Adding an I/O card to the Matlab installed boards list

If you have carried out a fresh install of the PCI-1711U card drivers, you will need to add the board to the list of installed boards that Matlab uses. The easiest way to do this is to carry out the following instructions.

In the Simulink CartControl window, identify the Feedback DAC block. Right click on this block and select *Look under mask* or press on the arrow in the bottom-left corner of the block. A new window will open showing the Simulink blocks that comprise the DAC block. Identify the *Analogue Output* block and double click on it. A dialog box will pop up asking whether you wish to add the board to the list of installed boards (Figure 22). Select yes.

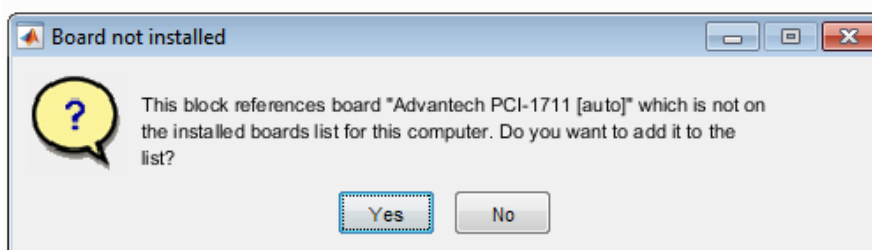


Figure 22 Adding an I/O Board to the Matlab Installed Boards List

The block parameter dialogue box for the PCI-1711U card will open. Cancel this dialogue then close the CartControl/Feedback DAC window. The PCI-1711U card is now correctly registered on the Matlab installed boards list.

Running a real-time model

You are now ready to run the CartControl application to test that the hardware and software has been installed correctly.

IMPORTANT

The analogue output of the PCI-1711U card is scaled from 0V to 5V and defaults to 0V output before it is initialized by a Simulink model. The motor driving the cart is scaled so that 0V gives full power to move the cart left, 2.5V gives zero power and 5V gives full power to move the cart to the right. Therefore, if the Digital Pendulum Controller is switched on and the green start button is pressed, the cart motor will be driven hard to the left into the mechanical end-stop. To prevent this, a valid model should be connected to the target first which will initialise the PCI-1711U card and set its output to 2.5V.

The following sequence is very important and should always be done, before executing a real-time model the first time since the startup of the computer, to avoid the condition described above. This only has to be done once after starting the computer. After ending a simulation, the cart is in a safe state.

1. Make sure that before you run the application, the Digital Pendulum Controller POWER switch is ON but the green START pushbutton is OFF (the green pushbutton lamp is not illuminated). If necessary, press the red STOP pushbutton.
2. Place the cart in the centre of the track (over the marker) and stop the pendulum from swinging.
3. Open Cart Control.mdl and start the simulation by pressing the play button. Wait until the model is build and already simulating (indicated by the word simulating in the bottom left corner and the scopes drawing data).
4. If the area around the pendulum is clear, you can press the green START pushbutton on the Digital Pendulum Controller unit.
5. When the simulation is complete, press the red STOP switch on the Digital Pendulum Controller.

Follow the sequence above and run the cart control model. During the simulation, the cart should move gently back and forth according to a sinusoidal reference signal. You can open the scopes to see how the position and angle signals vary but **note that the scope has to be opened before the application is started.**

More information on *CartControl.mdl* is given in the 'Control Experiments' manual.

Designing controllers

In this section, more information is given on the design of controllers.

It is best to start your own controller design by using an existing application and removing its content apart from two essential interface blocks (Figure 23), which are the Encoder input block (Cart Position/Pendulum Angle) and the DAC output block (Feedback DAC). By doing this, all of the simulation parameters that are needed for the Real Time application will be set.

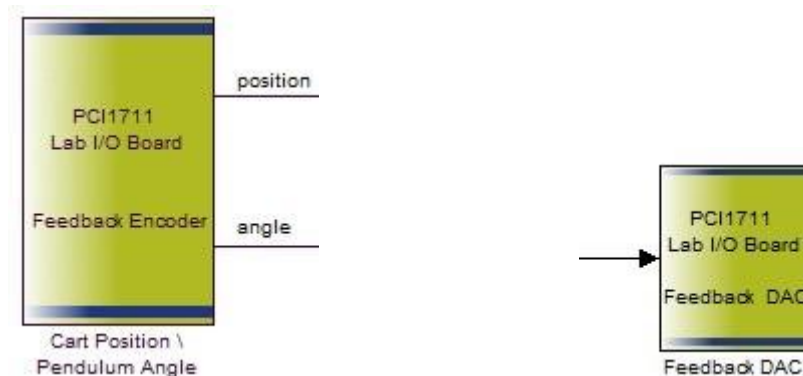


Figure 23 External Interface Blocks

The *Feedback encoder* block has two outputs. The upper output is the cart position in metres and the lower output is the pendulum angle measured in radians. You should not need to modify these blocks. Modifications may cause serious problems with control applications.

Whenever the simulation is started, the initial cart and pendulum position are set as a reference. Thus before starting the application the cart must be set at the zero point

(centre of the track) and the pendulum must hang vertically with as little movement as possible

It is best if you start by modifying the control parameters of existing applications or design your own simple algorithms to become familiar with the pendulum set-up.

Reading the encoders and setting the control signal

In this section the contents of the two blocks that interface with the external equipment, namely the '*Feedback encoder block*' and the '*Feedback DAC block*' are described (Figure 23).

PCI-1711U I/O Block – Feedback Encoder

- This is a universal Feedback block, which reads two 16-bit values from the encoder interface (two HCTL2016 ICs) and transforms them into relative position and relative angle.
- The block has two outputs which are both updated even if only one of them is used (connected) in the experiment. If one of the outputs is not used it is connected to a terminator.

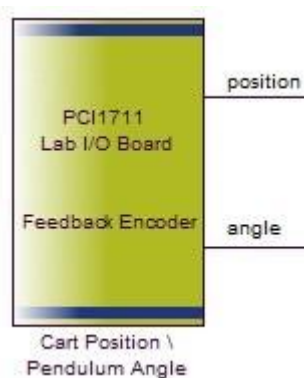


Figure 24 Encoder Interface Block

- There are three parameters for this block: sample time, channel one and channel two offsets. Channel one refers to the first encoder output and channel two to the second encoder output.
- By double clicking on this block we can change the above parameters (see Figure 25).

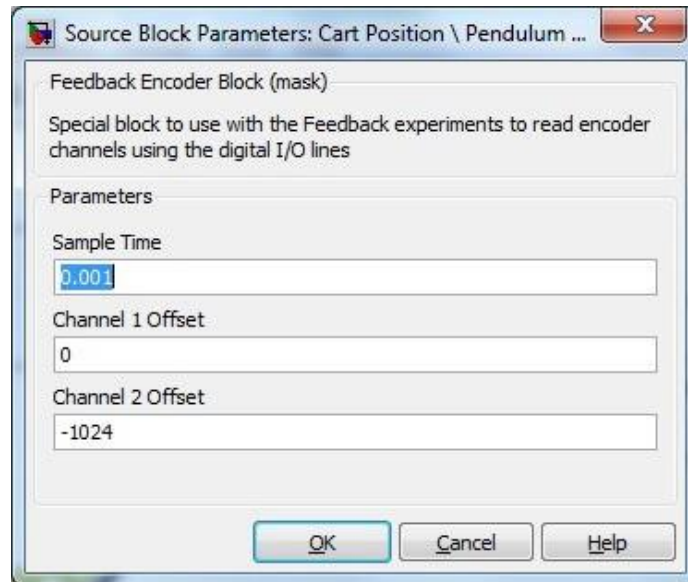


Figure 25 Feedback Encoder Block Parameters

- Choose “Look under Mask” to see the subsystems (Figure 26).

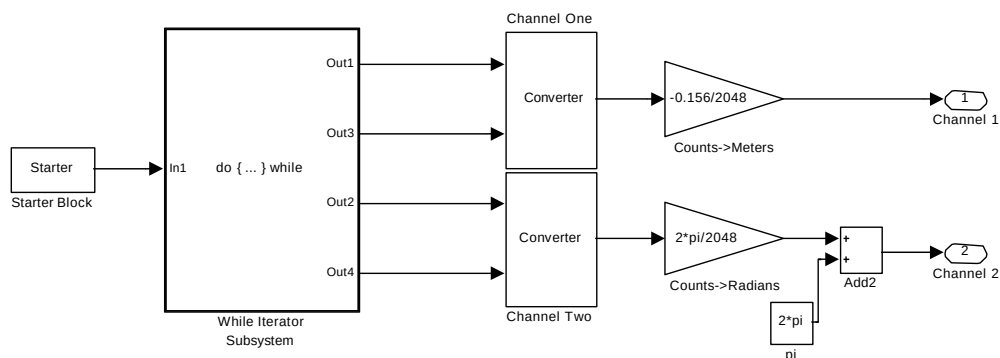


Figure 26 Feedback Encoder Block - Look Under Mask

- The “*While iterator subsystem*” reads low and high byte for the two encoders. These bytes are then converted into one 16-bit word for each encoder and we get two 16-bit words in the output of the block. We use the “*while iterator subsystem*” because we have to perform a lot of operations in the shortest time possible. These operations are necessary because every HCTL2016 encoder has an 8-bit output and reading a 16-bit word can only be done the following way:

- 1) prepare to read high byte: set !OE line to low and SEL line to low,
- 2) read high byte,
- 3) prepare to read low byte: set SEL line to high,
- 4) read low byte,
- 5) set !OE line to high (Figure 27).

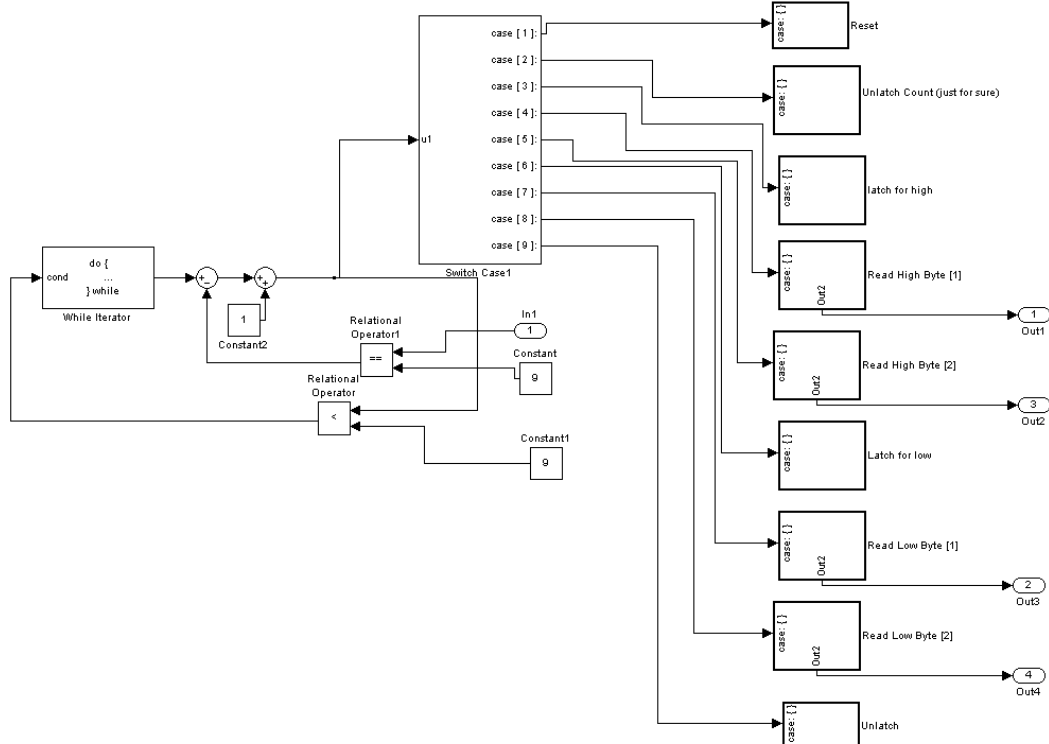


Figure 27 Feedback Encoder Block - Reading Values From Encoders

- Digital Output or Digital Input blocks are placed in the Action blocks presented in Figure 27.
- This block uses two S-functions: Starter and Converter. These are very simple functions and can be found in *C:\ProgramData\Feedback Instruments\33-936 Pendulum\src*.
- If the PCI-1711U card experiment is run for the first time we can double click on the Digital Output block and register the PCI-1711U card in Matlab environment (Figure 28).

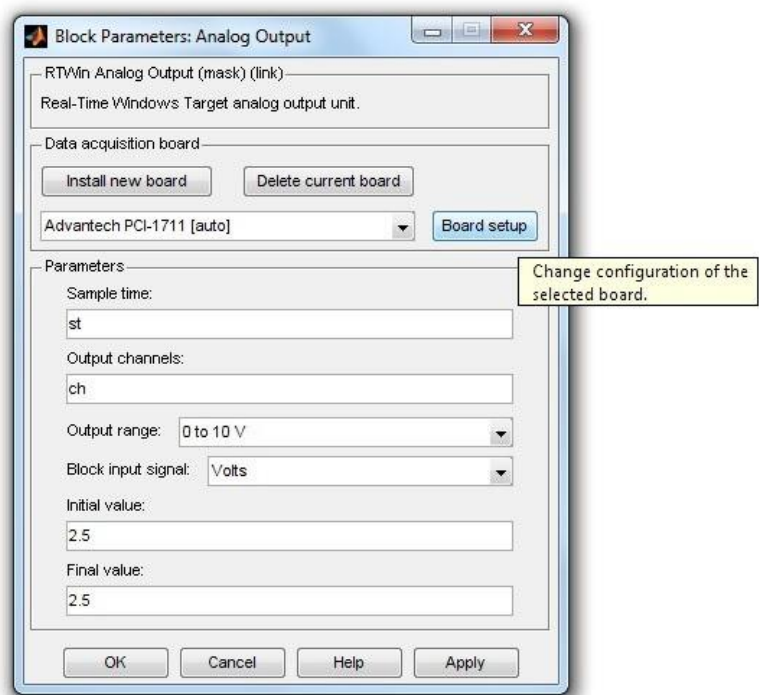


Figure 28 Selecting the I/O Card

PCI 1711U I/O Block – Feedback DAC

- This is a universal Feedback block which sends the input value to PCI-1711U analogue output.
- There is one input to this block. The block reads the value from this input and sends it to the analogue output of the PCI-1711U card (Figure 29).

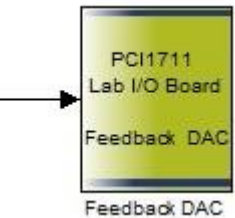


Figure 29 Feedback DAC Block

- There are two parameters available: sample time and output channel. We can choose between 1 or 2 for output channels. This is because the PCI-1711U has two analogue output channels. Channel 1 is connected to the cart motor.
- By double clicking on this block we can change the above parameters (Figure 30).

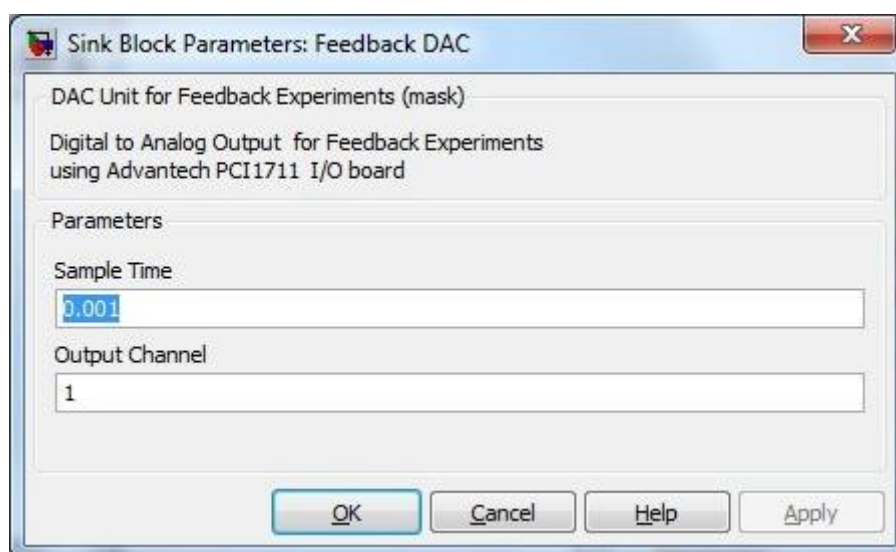


Figure 30 Feedback DAC Block Parameters

- The value, which is sent to the analogue output, is converted to the accepted range of 0V to 5V.
- Choosing “Look under Mask” will cause the subsystems to appear (Figure 31).

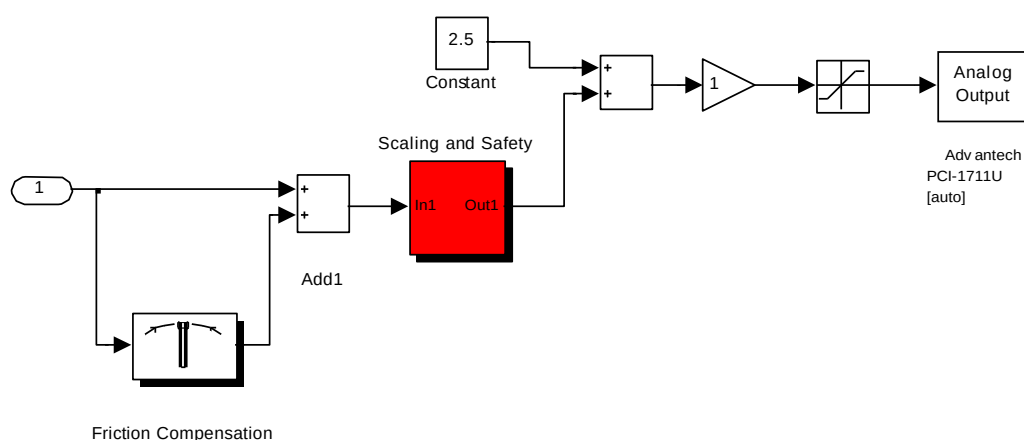


Figure 31 Feedback DAC Block - Under Mask

- If we run the PCI-1711U card experiment for the first time we can double click on the Analogue Output block and register the PCI-1711U card in the Matlab environment (Figure 32).

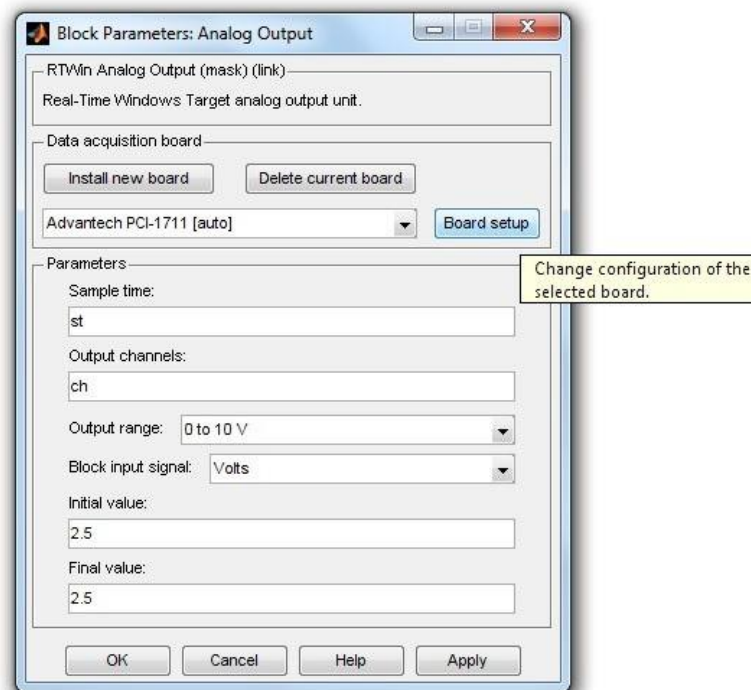


Figure 32 Feedback DAC Block - Selecting the I/O Card

Pendulum signals

- The Feedback Encoder block reads values from two encoders on the mechanical unit: cart position and pendulum angle. These two values must be scaled before we use them as controller inputs.
- Cart position should be set in metres and the scaling block converts from counts of the incremental encoder turning with the motor, to units of metres for the movement of the cart along the rails (Count->Metres gain block makes the conversion).
- Pendulum angle should be set in radians and the scaling block converts from count to radians. The offset is set to -1024 and this sets the pendulum reference position to Pi (Count->Radians gain block makes the conversion).
- The PID controller reads the values in metres and radians and gives the values already scaled to the required Volt units.

Troubleshooting

Controller runs weakly, cart motor seems to have little power

An important controller parameter is static friction (also called stiction). If your Pendulum mechanical unit is incorrectly adjusted, excessive stiction may result in poor performance. Run Exercise 3 and measure the stiction. A correctly set up mechanical unit will require approximately 0.2V to overcome stiction and cause the cart to move. Taking into account the scaling of the DAC (where 0V is full power left, 2.5V is zero and 5V is full power right), it should require around 2.7V to drive it to the right and 2.3V to the left (note that directions are viewed with the motor at the left of the track).

These results can be obtained using Exercise 3 or alternatively by driving the output of the DAC manually using the Advantech Device Manager programme, Test mode. The channel connected to the cart motor is labelled Channel 0 in the Advantech test programme.

CAUTION

When using Advantech Device Manager in Test mode, ensure that you set the output of Channel 0 to 2.5V (i.e. zero motor demand) before pressing the green START button on the Pendulum Controller Unit.

If you obtain stiction values significantly greater than $\pm 0.2V$ (i.e. outside the range 2.3V to 2.7V) the controller may not function correctly. First try to reduce stiction by checking the belt tension and adjusting if necessary. When the stiction is as low as you can achieve, rerun the stiction test to obtain new values. These new stiction values can be entered into the controller model as follows. The stiction component is incorporated into the DAC Output block of each controller. Look under the mask of the DAC Output block (Figure 33) Open the Friction Compensation block and enter the new values (Figure 34). You will need to rebuild the controller model.

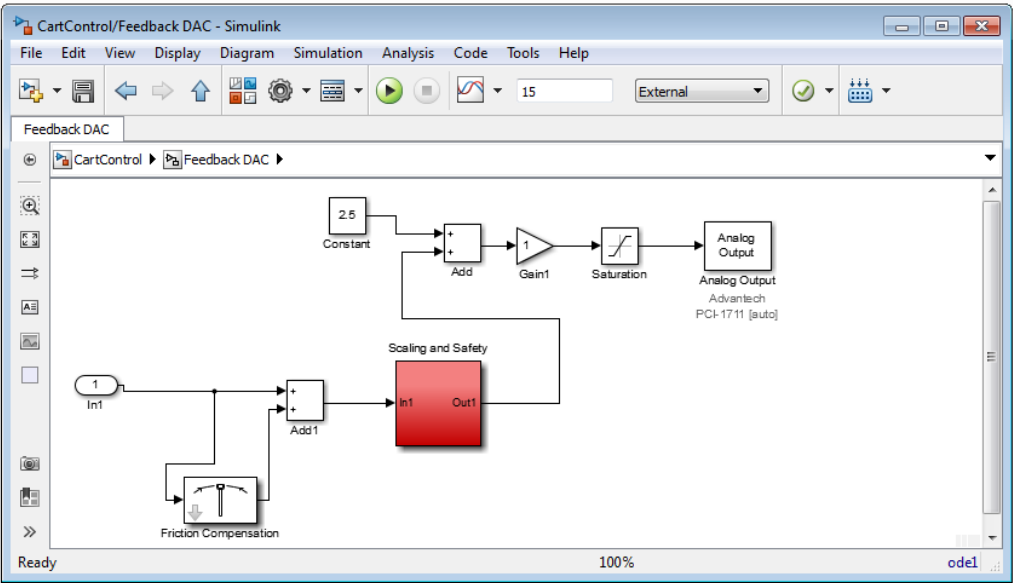


Figure 33 Look Under Mask of DAC Block

The screenshot shows the 'Function Block Parameters: Friction Compensation' dialog box. It has a 'linear (mask)' section and a 'Parameters' section. The 'Parameters' section contains two input fields: 'Friction Compensation Magnitude in Negative Direction' with a value of 1, and 'Friction Compensation Magnitude in Positive Direction' with a value of .1. At the bottom, there are buttons for 'OK', 'Cancel', 'Help', and 'Apply'.

Figure 34 Entering New Stiction Values

Masks have no background and show question marks

This usually happens when a model is opened directly from the examples folder. The shortcuts should be used for opening the models. If there are no shortcuts create one and give it an appropriate name. Right click on it and select properties. Replace the first line, labeled destination, in the shortcut tab with the following (all in one line, pressing the enter button is unnecessary):

```
"*Matlab location*\bin\matlab.exe" -r "open('C:\ProgramData\Feedback Instruments\33-936 Pendulum\Examples\PendulumModels.mdl');cd 'Examples'"
```

Replace *Matlab location* with the location where Matlab is installed. Example:

```
C:\Program Files\MATLAB\R2017b
```

Insert the following into the second line, labeled execute in:

```
"C:\ProgramData\Feedback Instruments\33-936 Pendulum"
```

Save the settings and open the shortcut. If the locations were correct the masks should load correctly. If the masks are still not displaying correctly please reinstall the models from the Feedback disk, as mentioned in the section "Pendulum software installation". Another workaround is to execute the startup.m script located in the last mentioned folder, before opening the models but without closing Matlab after executing the script.

Project files corrupted after crash during simulation or build stage

The Matlab error message shown in Figure 35 or similar may appear when Matlab crashed, while a model was being built or simulated.

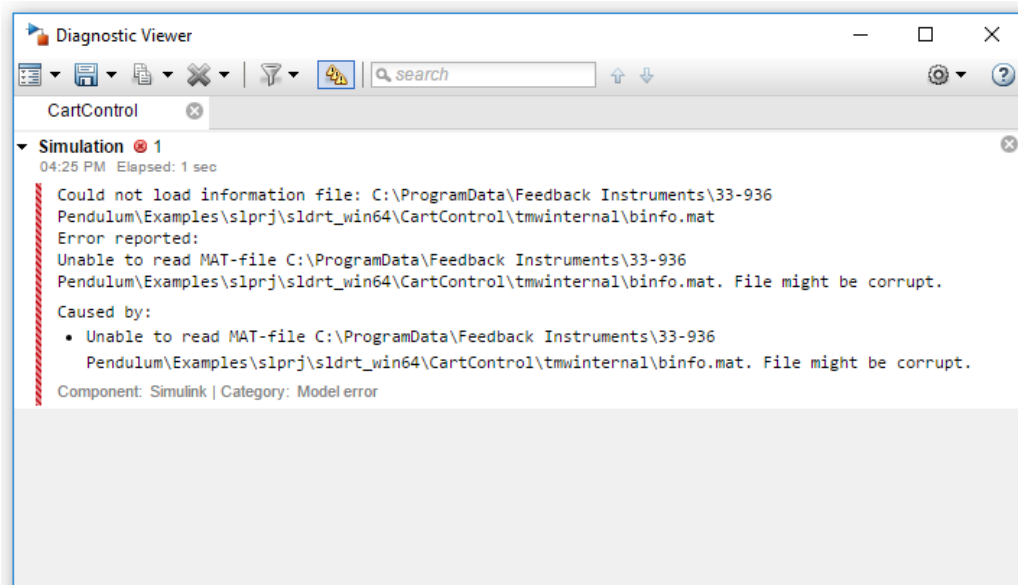


Figure 35 Matlab Error Message

To resolve the error open the examples folder and delete everything, except Real-time models, Simulation models and PendulumModels.mdl and try again. These files get generated when first building a model so it is safe to remove them. The examples folder should have the same contents like the examples backup folder.

Incorrect kernel version installed, when running a real-time simulation

An error message mentioning an incorrect kernel version may appear on several occasions, when a different version of Matlab is used. To check for the correct Kernel version enter:

`rtwho`

If a version mismatch is detected, close any instances of Simulink and uninstall the kernel with:

`sldrtkernel -uninstall`

and install the new kernel with:

`sldrtkernel -setup`

Note: The currently installed kernel version on the system is the same for all Matlab installations and every Matlab version needs its own kernel version. This means, that if you have to simulate Simulink real-time models with different versions of Matlab on the same computer, you will always have to do these steps when switching between the different Matlab versions.

Unresponsive Windows, freeze or crash on simulation

If you experience freezing, crashing or Windows becoming weirdly unresponsive when trying to start a real-time simulation try changing some of the BIOS settings of the computer:

- Turning **off** an option called **VT-d** (Intel Virtualization Technology for Directed I/O)
- Turning **on** an option called **HPET** (High Performance/Prevision Event Timer)
- Turning **off** an option called **C1E** or enhanced halt. This setting is known to cause problems with the real-time kernel.
- Turning **off** other energy saving measures might also be worth a try. Keep in mind that turning off these settings increases energy consumption.

If you still experience problems, try updating (downloading and reinstalling) you current version of Matlab, a different version of Matlab or a different Windows operating system (within the specs mentioned at the beginning of this document).